

Calculations of Proportion of Total Effect Mediated (P_M) and Absolute P_M in Complex Mediation Models

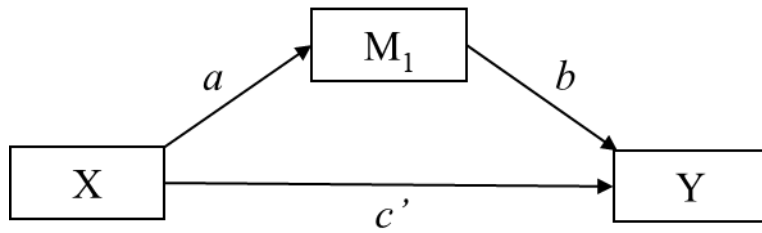
Research Goal	P_M Equation	<i>Absolute P_M</i> Equation
(1) Determine the mediator's effect as influenced by the moderator z (e.g., Model 2).	$\frac{(ab)z}{[(ab)z + c']}$	$\frac{ (ab)z }{ (ab)z + c' }$
(2) Isolate the effect of a single mediator (M_1) among multiple mediators (e.g., Model 3)	$\frac{(a_1b_1)}{(a_1b_1 + a_2b_2 + c')}$	$\frac{ a_1b_1 }{ a_1b_1 + a_2b_2 + c' }$
(3) Determine the total mediated effect among multiple mediators (M_1+M_2) (e.g., Model 3)	$\frac{(a_1b_1 + a_2b_2)}{(a_1b_1 + a_2b_2 + c')}$	$\frac{ a_1b_1 + a_2b_2 }{ a_1b_1 + a_2b_2 + c' }$
(4) Isolate the effect of a single mediator (M_2) in a serial (i.e., interlinked) mediation scenario (e.g., Model 4)	$\frac{(a_1a_3b_2 + a_2b_2)}{(a_1a_3b_2 + a_1b_1 + a_2b_2 + c')}$	$\frac{ a_1a_3b_2 + a_2b_2 }{ a_1a_3b_2 + a_1b_1 + a_2b_2 + c' }$
(5) Determine the total mediated effect (M_1+M_2) in a serial (i.e., interlinked) mediation scenario (e.g., Model 4)	$\frac{(a_1a_3b_2 + a_1b_1 + a_2b_2)}{(a_1a_3b_2 + a_1b_1 + a_2b_2 + c')}$	$\frac{ a_1a_3b_2 + a_1b_1 + a_2b_2 }{ a_1a_3b_2 + a_1b_1 + a_2b_2 + c' }$

Note. All path coefficients (e.g., a , b , a_1 , b_1 , etc.) correspond to the models in Figure 1.

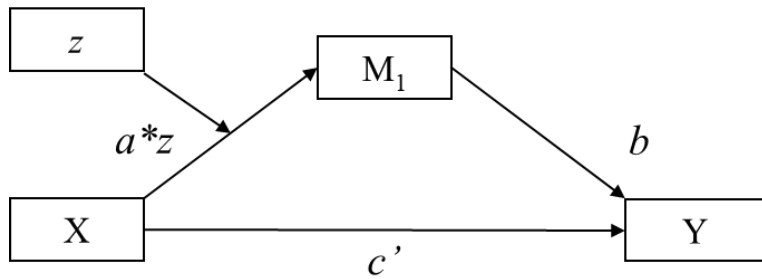
Figure 1

Illustrative Examples of Mediation Models

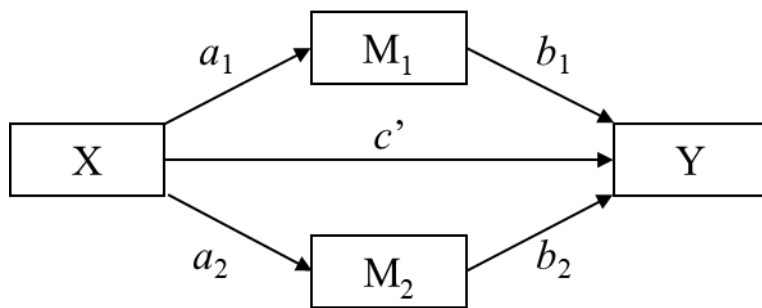
Model 1: Simple Mediation



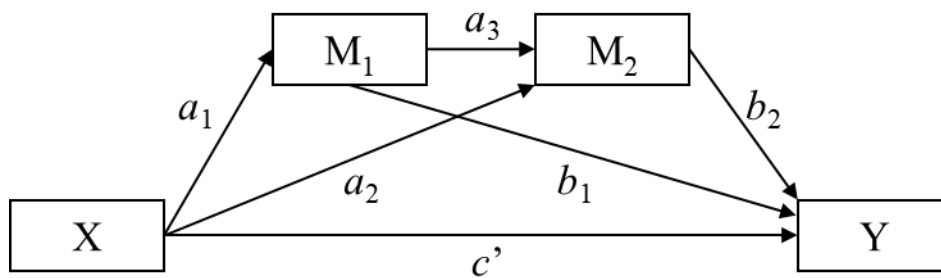
Model 2: First Stage Moderated Mediation



Model 3: Multiple Mediation



Model 4: Serial Mediation



Note: In all models, c represents the total effect of X on Y.